

Skills Summary

Solving Logarithmic Equations with Domain Checks

Use this reference while completing your worksheet or when studying.

Skill 1 — Converting a logarithm to exponential form

Definition. For $b > 0$, $b \neq 1$: $\log_b A = c$ means $A = b^c$.

Domain rule. The argument A must be strictly positive. Write that condition down *before* solving — it is what tells you which answers to keep.

Example: Solve $\log_2(x - 5) = 4$.

Step 1. One logarithm, a number on the other side — convert straight to exponential form, then check the argument stays positive.

Step 2. $x - 5 = 2^4 = 16$, so $x = 21$; then $21 - 5 = 16 > 0$.

$$\Rightarrow x = 21$$

Try it: Solve $\log_3(x + 2) = 3$.

check: $25 = x$

Skill 2 — Equating arguments of matching logarithms

One-to-one property. $\log_b A = \log_b B$ means $A = B$, provided $A > 0$ and $B > 0$.

A candidate that makes either argument negative is extraneous, even though it satisfies $A = B$.

Example: Solve $\log_7(4x - 1) = \log_7(x + 8)$.

Step 1. The bases match and each side is a single logarithm, so the one-to-one property lets me drop both logs at once.

Step 2. $4x - 1 = x + 8$, so $3x = 9$; at $x = 3$ the arguments are 11 and 11, both positive.

$$\Rightarrow x = 3$$

Try it: Solve $\log_2(3x + 4) = \log_2(x + 12)$.

check: $4 = x$

Skill 3 — Condensing two logarithms, then solving

Product: $\log_b M + \log_b N = \log_b(MN)$. **Quotient:** $\log_b M - \log_b N = \log_b\left(\frac{M}{N}\right)$.

Condensing usually produces a quadratic, so expect two candidates and expect the domain to reject one.

Example: Solve $\log_2 x + \log_2(x - 2) = 3$.

Step 1. Two logarithms with the same base added together — condense with the product rule first, since the domain needs $x > 2$.

Step 2. $x(x - 2) = 2^3 = 8$, so $x^2 - 2x - 8 = 0$ and $(x - 4)(x + 2) = 0$; $x = -2$ fails $x > 2$.

$$\Rightarrow x = 4$$

Try it: Solve $\log_5 x + \log_5(x - 4) = 1$.

check: $x = 5$